

Chanuga Liyanage

Mechanical Engineering Undergraduate Specializing in Mechatronics

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PERSONAL PROFILE

Final-year Mechanical Engineering undergraduate specializing in Mechatronics at the University of Peradeniya with a strong foundation in mechanical design, automation, embedded systems, and computational engineering. Experienced in applying engineering principles, simulation tools, and data-driven methods to solve real-world problems through academic, industrial, and research projects. A motivated and adaptable individual with strong analytical and problem-solving skills, committed to contributing effectively in a professional engineering environment while continuously learning and developing technical expertise.

EDUCATION

University of Peradeniya 2022 – 2026

B.Sc. Engineering (Hons) in Mechanical Engineering, Specialized in Mechatronics GPA: 3.60 / 4.00

EXPERIENCE

KSPA Panadura | *Mechanical Engineering Intern* September 2025 – November 2025

- Gained hands-on experience in corrugated manufacturing processes and industrial machinery, developing an understanding of machine mechanisms, production workflows, maintenance, troubleshooting, and the application of cutting tools and automation components in manufacturing operations.

Vega Innovations | *Battery System Engineering Intern* July 2024 – October 2024

- Gained hands-on experience in electric vehicle technology and mechanical systems, contributing to design and development activities, technical operations, and project management while enhancing knowledge of EVs and industrial machinery.

PROJECTS

RideTwin: AI-Based Digital Twin for Vehicle Ride Comfort Analysis Final Year Research

- Developed a physics-based digital twin of a quarter-car suspension using MATLAB/Simulink, applied ISO 2631 ride comfort evaluation and signal processing, and built machine learning models for ride comfort prediction and suspension parameter optimization.

Hybrid Electric Vehicle (HEV) Controller Design Group Project

- Designed and implemented a microcontroller-based control system for a series hybrid electric vehicle prototype, validated battery and motor interaction through MATLAB/Simulink simulations, and developed a scaled-down working model demonstrating real-time power source switching and speed control.

Cell Compression Machine Design Intern

- Designed a pneumatic cell compression machine for LiFePO₄ battery cells, developing skills in pneumatic system design, force calculations, component selection, and mechanical design to improve battery cycle life and structural integrity.

V6 Engine Modeling and Assembly using SolidWorks

Individual

- Modeled and assembled a functional 3D V6 engine in SolidWorks, gaining a comprehensive understanding of the design and operation of individual engine components.

AWARDS

- First Place at the Skyward Challenge organized by KDU IESL at the Techno 2024 Exhibition.
- First Place at Gravity Games 2024 organized by SEDS Earth in collaboration with SEDS Sri Lanka.
- First Runners-up at the Aero Day Quiz Competition organized by Hybrid Education in collaboration with RC Falcons.

TRAINING & CERTIFICATIONS

Industrial Automation Workshop (PLC and SCADA)

July 2024

- Conducted by Eng. Sugath Pradeep, Head of the Technical Industry Business, IET University of Peradeniya. Gained practical knowledge of PLC and SCADA technologies through interactive demonstrations and real-world industrial applications.

Supervised Machine Learning: Regression and Classification

June 2024

DeepLearning.AI & Stanford University (Coursera)

- Developed supervised machine learning models in Python using NumPy and Scikit-learn for regression and binary classification tasks, including linear and logistic regression.

Advanced Learning Algorithms

June 2024

DeepLearning.AI & Stanford University (Coursera)

- Built and trained neural network models using TensorFlow for multi-class classification and applied machine learning best practices to improve model generalization and real-world performance.

VOLUNTEERING

Committee Member, Automotive Society, University of Peradeniya

2025

Member, Students for the Exploration and Development of Space (SEDS), University of Peradeniya

Present

Member, AeroGenZ Society, University of Peradeniya

Present

EXTRACURRICULAR ACTIVITIES

Sports

- **Weightlifting**
 - 2nd Place, Inter-Faculty Freshers' Weightlifting Championship 2023

REFERENCES

Prof. Primal Fernando

PhD Specialization in Thermodynamics and Refrigeration

Head of the Department

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Prof. Asanga Ratnaweera

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